

# Prompt Engineering and Large Language Models

## Midterm Examination

Weeks 1–7 · 25 questions  $\times$  4 points = 100 points · 40% of the course grade

Name: \_\_\_\_\_ Student No: \_\_\_\_\_ Date: \_\_\_\_\_

**Instructions.** This paper contains 25 multiple-choice questions. Select the **single best** answer for each. Each question is worth equal marks. Time allowed: 50 minutes. No notes or electronic devices.

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1. Which of the following tasks is **generative**?
  - A) Ranking job applications by predicted suitability
  - B) Flagging a transaction as possibly fraudulent
  - C) Drafting a discharge letter for a patient's family
  - D) Detecting whether an image contains a tumour
2. Which statement about the term **prompt** is correct?
  - A) It refers only to the text you type
  - B) It includes attached documents and images
  - C) It excludes any instruction added by the product
  - D) It refers to the model's response
3. A model answers a question on a topic for which it has very little training material. What happens to the confidence expressed in its answer?
  - A) It falls in proportion to the thinness of the coverage
  - B) It stays essentially the same
  - C) It is reported to the user as a percentage
  - D) The model refuses rather than answering
4. An AI summary of a report is accurate throughout but omits the single most significant finding. Which failure signature is this?
  - A) Fabrication
  - B) Displacement
  - C) Omission
  - D) Silent failure
5. ELIZA (1966) contained:
  - A) A small knowledge base of medical facts
  - B) A simple neural network
  - C) Keyword matching and sentence reflection, with no knowledge or memory
  - D) A statistical model of English grammar
6. What is the "ELIZA effect"?
  - A) The tendency of early chatbots to crash under load

- B) The attribution of understanding to a system that produces plausible language
  - C) The improvement of output when a persona is specified
  - D) The degradation of models trained on their own output
7. The AI winters were primarily caused by:
- A) Technical failure of the underlying methods
  - B) The gap between what was promised and what was delivered
  - C) Shortages of computing hardware
  - D) Legal restrictions on research
8. A long agglutinated Turkish word is likely to be split into several tokens because:
- A) Turkish grammar is irregular
  - B) The complete assembled word appeared rarely in training text, though its pieces are common
  - C) Turkish uses diacritics
  - D) Tokenizers cannot process non-English alphabets
9. Which is **not** a consequence of Turkish costing more tokens than English?
- A) The context window fills faster
  - B) Metered usage costs more for the same content
  - C) The model refuses Turkish input
  - D) Performance is often weaker due to thinner training coverage
10. The model's core operation is best described as:
- A) Looking up the most relevant stored fact
  - B) Producing a probability distribution over possible next tokens
  - C) Applying a decision tree written by its developers
  - D) Searching a database of documents
11. Which statement about the model's internal state is correct?
- A) It knows when it is guessing but does not report it
  - B) It performs the same operation whether the pattern is strong or weak
  - C) It has a separate mode for uncertain answers
  - D) It calculates a confidence score for each answer
12. Why can these systems not simply reply "I don't know" whenever appropriate?
- A) The phrase is filtered out by safety systems
  - B) It is a token sequence competing with all others, and questions in training text are usually followed by answers
  - C) The model is instructed never to refuse
  - D) It would use too many tokens
13. The context window contains all of the following **except**:
- A) Your current message
  - B) Documents you attached
  - C) Hidden instructions added by the product

- D) Everything the model learned during training
14. When a document exceeds the context window, the most dangerous behaviour is:
- A) The tool refuses the input with an explicit error
  - B) Part of the material is silently dropped and a confident answer is produced anyway
  - C) The response is visibly truncated
  - D) The model asks you to shorten the document
15. Which is **not** one of the four levers?
- A) Context
  - B) Task
  - C) Format
  - D) Temperature
16. Output is accurate and complete but reads like nothing you would ever send. Which lever should you pull?
- A) Context
  - B) Task
  - C) Format
  - D) Examples
17. Which is the strongest reason to split “summarise, translate, and identify risks” into three separate turns?
- A) It uses fewer tokens overall
  - B) Each job receives fuller attention, and you can check each result before building on it
  - C) The model cannot perform three tasks
  - D) Translation requires a different model
18. Someone adds a phrase to a prompt, the output improves, and they conclude the phrase works. The strongest objection is:
- A) The phrase may be too long
  - B) Output varies between runs, so the improvement may be unrelated to the change
  - C) The model may not have read the phrase
  - D) The phrase may be in the wrong language
19. Which instruction converts an unverifiable claim into a checkable one?
- A) “Be accurate and thorough”
  - B) “Only use reliable sources”
  - C) “After each claim, quote the sentence that supports it”
  - D) “Think carefully before answering”
20. In the three dimensions of register, **tenor** refers to:
- A) The subject area and its vocabulary density
  - B) The relationship between writer and reader
  - C) The channel of communication
  - D) The emotional warmth of the text

21. Which specification is most likely to produce reproducible style?
- A) “Make it professional and friendly”
  - B) “Everyday vocabulary with no unglossed medical terms; warm but not casual; written to be read aloud”
  - C) “Use an appropriate tone for the audience”
  - D) “Write it well”
22. Why does a stated fallback such as “write NOT STATED” reduce invented content?
- A) It reduces the model’s temperature
  - B) It makes the honest answer a probable continuation rather than an unlikely one
  - C) It prevents the model from accessing training data
  - D) It shortens the output
23. A generated browser tool produces a subtly wrong calculation but has a clean, working interface. Why is this particularly dangerous?
- A) The error is intermittent and therefore hard to reproduce
  - B) The error is consistent, which makes the tool appear reliable
  - C) The tool will eventually stop working
  - D) The interface will change without warning
24. Grounding a system in your own documents changes:
- A) What the model knows
  - B) Where the answer comes from
  - C) Whether the model checks the truth of its output
  - D) The model’s training data
25. An assistant grounded in your documents produces an answer with no supporting quotation. The most likely explanation is:
- A) It over-read the passage
  - B) It retrieved the wrong passage
  - C) It answered from memory
  - D) The context window overflowed